Отчёт по лабораторной работе №7

Имитационное моделирование

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1 Цель работы

Освоить дискретно-событийное моделирование в Julia на примерах модели M/M/c и модели Росса, провести параметрические исследования.

2 Задание

— Создать рабочий каталог для кода. — Установить необходимые пакеты. — Выполнить предложенный код. — Преобразовать код в литературный стиль. — Сгенерировать из литературного кода: — чистый код; — jupyter notebook; — документацию в формате Quarto. — Выполнить код из jupyter notebook. — Интегрировать документацию в формате Quarto в отчёт. — Добавить в код в литературном стиле вычисление для набора параметров. — Сгенерировать из литературного кода с параметрами: — чистый код; — jupyter notebook; — документацию в формате Quarto. — Выполнить код из jupyter notebook с параметрами. — Интегрировать документацию с параметрами в формате Quarto в отчёт.

3 Теоретическое введение

3.1 Модель M/M/c

Система массового обслуживания с c серверами:

- **M** (Markovian) — входящий поток пуассоновский с интенсивностью λ (интервалы прибытия $\sim \text{Exp}(1/\lambda)$).
- **M** — время обслуживания экспоненциальное с интенсивностью μ ($\sim \text{Exp}(1/\mu)$).
- **c** — число параллельных серверов.

Загрузка системы: $\rho = \lambda/(c\mu)$. Условие стационарности $\rho < 1$.

3.2 Модель Росса

Система с N работающими машинами и S запасными:

- Один ремонтник (обслуживает по одной машине).
- Наработка до отказа одной машины $\sim \text{Exp}(\lambda)$.
- Время ремонта $\sim \text{Exp}(\mu)$.
- При отказе машины одна запасная **немедленно** берётся из резерва.
- Если резерв **пуст** — система **падает** (crash).
- После ремонта машина **возвращается** в резерв.

Требуется найти **среднее время до падения** $E[T]$.

3.3 Дискретно-событийное моделирование

DES (Discrete-Event Simulation) — моделирование систем, состояние которых меняется **мгновенно** в определённые моменты времени («события»). В Julia реализуется пакетом `ConcurrentSim.jl` через макросы `@resumable` и `@yield`.

4 Выполнение лабораторной работы

4.1 Создание рабочего пространства

DrWatson-проект создан в каталоге `labs/lab07/project` (рис. Рисунок 4.1).

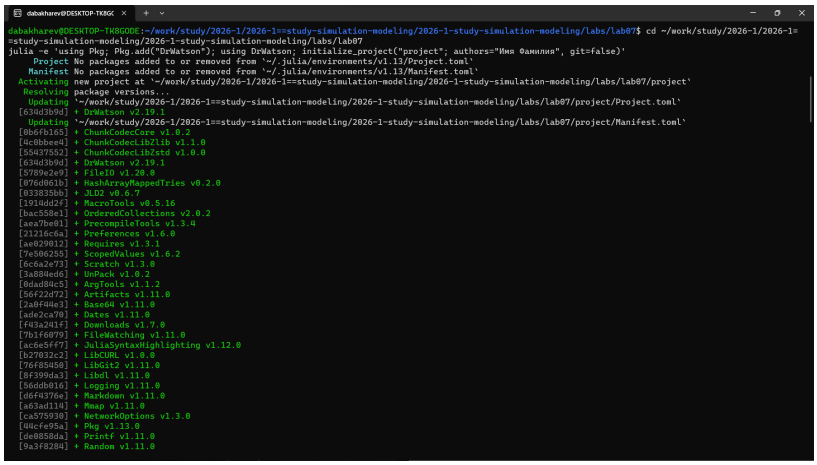


Рисунок 4.1: Создание DrWatson-проекта

4.2 Установка пакетов

Создаю файл `add_packages.jl` для установки пакетов `ConcurrentSim`, `ResumableFunctions`, `StableRNGs`, `Distributions` (рис. Рисунок 4.2).



Рисунок 4.2: Файл add_packages.jl

Запускаю его — все пакеты устанавливаются успешно (рис. Рисунок 4.3).

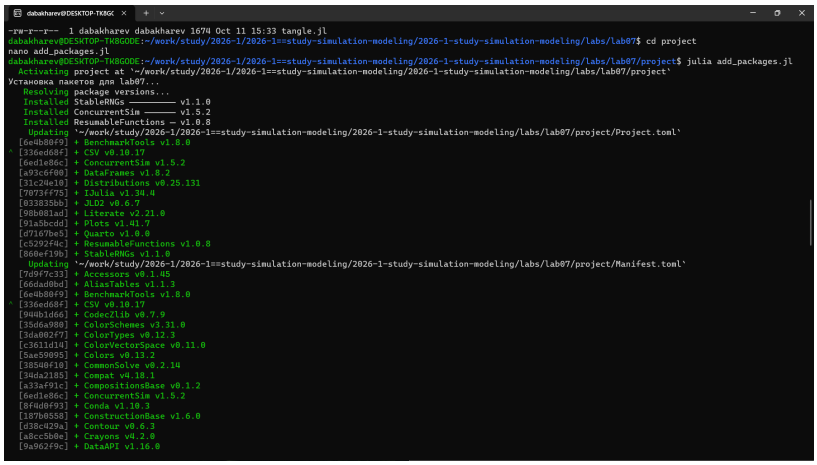


Рисунок 4.3: Установка пакетов Julia

4.3 Модель M/M/c

Создаю скрипт `scripts/mmc_queue.jl` с моделью M/M/c (число серверов $c = 2$, $\lambda = 0.9$, $\mu = 0.5$). Запускаю его (рис. Рисунок 4.4).

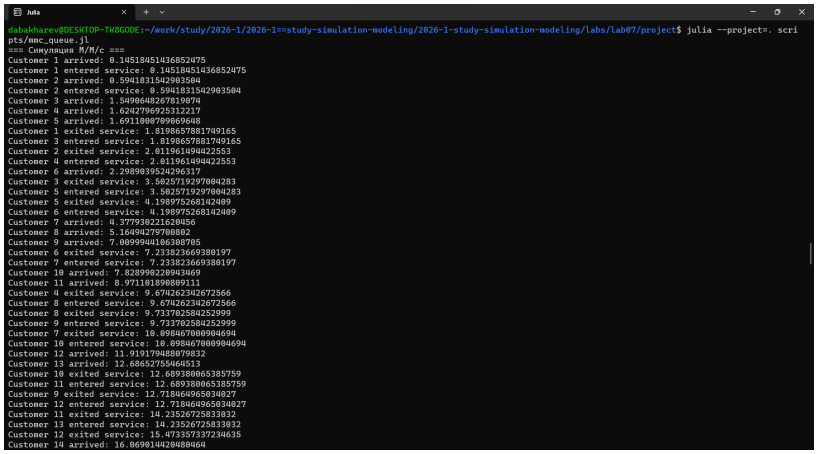


Рисунок 4.4: Запуск модели M/M/c

В выводе видны события: `Customer N arrived`, `entered service`, `exited service`. События обрабатываются **в порядке времени**.

4.4 Модель Росса

Создаю скрипт `scripts/ross_model.jl` для модели Росса ($N = 10$, $S = 3$, $\lambda = 100$, $\mu = 1$). Запускаю его (рис. Рисунок 4.5).

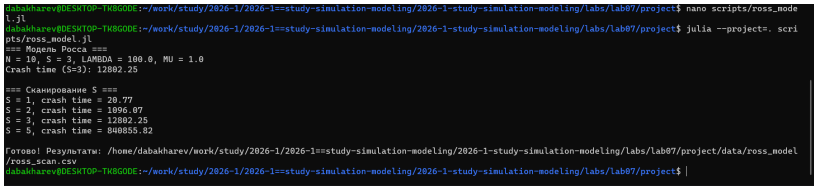


Рисунок 4.5: Запуск модели Росса

Результаты сканирования по S:

S	Среднее время до отказа
1	20.77
2	1096.07
3	12802.25
5	840855.82

Чем больше запасных машин, тем дольше система работает — это иллюстрирует важность резервирования.

4.5 Генерация производных форматов

Запускаю `tangle.jl` для скриптов `mmc_queue.jl` и `ross_model.jl`. Генерируются `.qmd`, `.ipynb` и чистый `.jl` (рис. Рисунок 4.6).



Рисунок 4.6: Генерация производных форматов

4.6 Результаты M/M/c

График зависимости среднего времени ожидания и обслуживания от λ (рис. Рисунок 4.7).

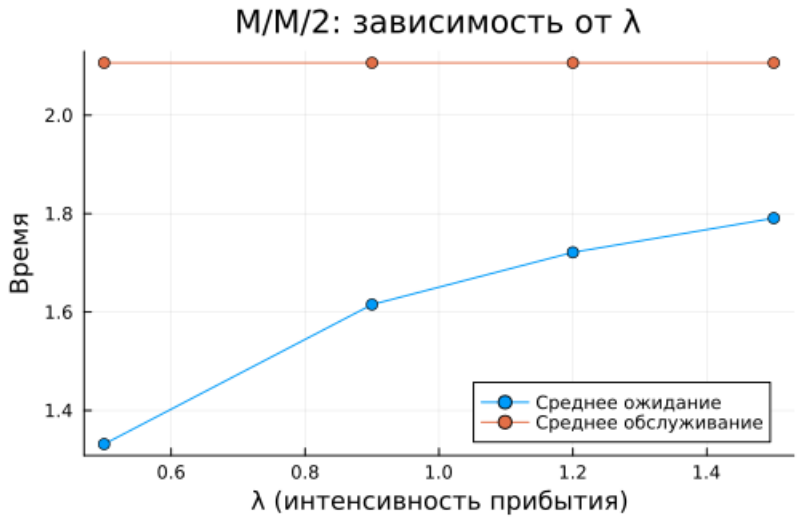


Рисунок 4.7: Результаты M/M/c

Видно, что с ростом λ среднее время ожидания **растёт**.

4.7 Результаты модели Росса

График зависимости времени до отказа от числа запасных машин S (рис. Рисунок 4.8).

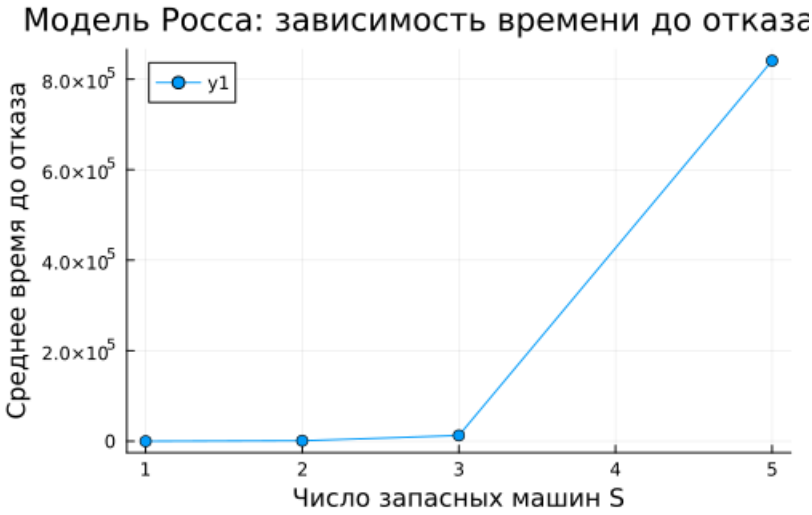


Рисунок 4.8: Результаты модели Росса

Кривая **резко возрастает** — экспоненциальная зависимость от S.

4.8 Листинги из Julia

5 Модель M/M/c (система массового обслуживания)

Стохастическая симуляция через дискретно-событийное моделирование (ConcurrentSim).

```
import Pkg
Pkg.activate("/home/dabakharev/work/study/2026-1/2026-1==study-simu
using DrWatson

using StableRNGs
using Distributions
using ConcurrentSim
using ResumableFunctions
using DataFrames, CSV, Plots, Statistics

ENV["GKSwstype"] = "100"

script_name = "mmc_queue"
mkpath(plotsdir(script_name))
mkpath(datadir(script_name))
```

```
  Activating project at `~/work/study/2026-1/2026-1==study-simulation
```

"/home/dabakharev/work/study/2026-1/2026-1==study-simulation-modeli

5.1 Параметры симуляции

```
rng = StableRNG(123)
num_customers = 100           # □□□□□ □□□□□ □□□□□□□□
num_servers = 2               # □□□□□ □□□□□□□□
mu = 1.0 / 2                  # □□□□□□□□□□□□□ □□□□□□□□□□□□
lam = 0.9                     # □□□□□□□□□□□□□ □□□□□□□□
arrival_dist = Exponential(1 / lam)   # □□□□□□□□□□□□□ □□□□□□□□□□ □□□□□□□□
service_dist = Exponential(1 / mu)    # □□□□□□□□□□□□□ □□□□□□□□ □□□□□□□□□□□□
```

```
Exponential{Float64}(□=2.0)
```

5.2 Поведение клиента

```
@resumable function customer(env::Environment, server::Resource, id:
                             t_a::Float64, d_s::Distribution)
    @yield timeout(env, t_a)
    arrival_time = now(env)
    println("Customer $id arrived: ", arrival_time)

    @yield request(server)
    service_start = now(env)
   println("Customer $id entered service: ", service_start)

    service_time = rand(rng, d_s)
    @yield timeout(env, service_time)
```

```
    @yield unlock(server)
    service_end = now(env)
    println("Customer $id exited service: ", service_end)
end
```

```
customer (generic function with 1 method)
```

5.3 Функция запуска симуляции

```
function setup_and_run()
    sim = Simulation()
    server = Resource(sim, num_servers)

    arrival_time = 0.0
    for i in 1:num_customers
        arrival_time += rand(rng, arrival_dist)
     @process customer(sim, server, i, arrival_time, service_dist)
    end
    run(sim)
end

println("=== □□□□□□□□ M/M/c ===")
setup_and_run()
```

```
=== □□□□□□□□ M/M/c ===
Customer 1 arrived: 0.14518451436852475
Customer 1 entered service: 0.14518451436852475
Customer 2 arrived: 0.5941831542903504
```

```
Customer 2 entered service: 0.5941831542903504
Customer 3 arrived: 1.5490648267819074
Customer 4 arrived: 1.6242796925312217
Customer 5 arrived: 1.6911000709069648
Customer 1 exited service: 1.8198657881749165
Customer 3 entered service: 1.8198657881749165
Customer 2 exited service: 2.011961494422553
Customer 4 entered service: 2.011961494422553
Customer 6 arrived: 2.2989039524296317
Customer 3 exited service: 3.5025719297004283
Customer 5 entered service: 3.5025719297004283
Customer 5 exited service: 4.198975268142409
Customer 6 entered service: 4.198975268142409
Customer 7 arrived: 4.377930221620456
Customer 8 arrived: 5.16494279700802
Customer 9 arrived: 7.0099944106308705
Customer 6 exited service: 7.233823669380197
Customer 7 entered service: 7.233823669380197
Customer 10 arrived: 7.828990220943469
Customer 11 arrived: 8.971101890809111
Customer 4 exited service: 9.674262342672566
Customer 8 entered service: 9.674262342672566
Customer 8 exited service: 9.733702584252999
Customer 9 entered service: 9.733702584252999
Customer 7 exited service: 10.098467000904694
Customer 10 entered service: 10.098467000904694
Customer 12 arrived: 11.919179488079832
Customer 13 arrived: 12.68652755464513
Customer 10 exited service: 12.689380065385759
```

```
Customer 11 entered service: 12.689380065385759
Customer 9 exited service: 12.718464965034027
Customer 12 entered service: 12.718464965034027
Customer 11 exited service: 14.23526725833032
Customer 13 entered service: 14.23526725833032
Customer 12 exited service: 15.473357337234635
Customer 14 arrived: 16.069014420480464
Customer 14 entered service: 16.069014420480464
Customer 14 exited service: 16.32875421109624
Customer 13 exited service: 16.482625343974227
Customer 15 arrived: 18.14316642520691
Customer 15 entered service: 18.14316642520691
Customer 15 exited service: 18.845530557075612
Customer 16 arrived: 22.458520156865283
Customer 16 entered service: 22.458520156865283
Customer 16 exited service: 25.002252321560913
Customer 17 arrived: 25.45645888639873
Customer 17 entered service: 25.45645888639873
Customer 18 arrived: 26.338997953649585
Customer 18 entered service: 26.338997953649585
Customer 18 exited service: 26.350799415378106
Customer 19 arrived: 26.85595179780316
Customer 19 entered service: 26.85595179780316
Customer 19 exited service: 26.868476149244078
Customer 20 arrived: 27.57174714247525
Customer 20 entered service: 27.57174714247525
Customer 17 exited service: 27.912706919394097
Customer 21 arrived: 29.066330171566218
Customer 21 entered service: 29.066330171566218
```

```
Customer 20 exited service: 29.71916951346357
Customer 22 arrived: 29.973500123225932
Customer 22 entered service: 29.973500123225932
Customer 22 exited service: 30.062708620746623
Customer 23 arrived: 32.606843734849136
Customer 23 entered service: 32.606843734849136
Customer 23 exited service: 33.36629309490528
Customer 24 arrived: 33.50064850325357
Customer 24 entered service: 33.50064850325357
Customer 21 exited service: 34.34801404289969
Customer 25 arrived: 34.547861690473496
Customer 25 entered service: 34.547861690473496
Customer 25 exited service: 34.66999919462739
Customer 26 arrived: 35.02482413816678
Customer 26 entered service: 35.02482413816678
Customer 24 exited service: 35.589301906593
Customer 26 exited service: 36.54324299243115
Customer 27 arrived: 36.60907362518617
Customer 27 entered service: 36.60907362518617
Customer 28 arrived: 37.028705340760354
Customer 28 entered service: 37.028705340760354
Customer 27 exited service: 37.91458137337213
Customer 29 arrived: 39.526657099378426
Customer 29 entered service: 39.526657099378426
Customer 28 exited service: 39.76758125651681
Customer 30 arrived: 39.83441009316409
Customer 30 entered service: 39.83441009316409
Customer 31 arrived: 40.44787524447751
Customer 32 arrived: 41.075350408403665
```

```
Customer 33 arrived: 41.62165226663721
Customer 34 arrived: 42.259181189225714
Customer 35 arrived: 43.2109131723922
Customer 29 exited service: 43.728988543326565
Customer 31 entered service: 43.728988543326565
Customer 36 arrived: 45.35268621772801
Customer 30 exited service: 45.76624408293789
Customer 32 entered service: 45.76624408293789
Customer 31 exited service: 45.8246116638791
Customer 33 entered service: 45.8246116638791
Customer 37 arrived: 46.14560779296951
Customer 33 exited service: 46.35014258445585
Customer 34 entered service: 46.35014258445585
Customer 32 exited service: 47.076030276412446
Customer 35 entered service: 47.076030276412446
Customer 35 exited service: 47.5643764242456
Customer 36 entered service: 47.5643764242456
Customer 34 exited service: 47.84368765556911
Customer 37 entered service: 47.84368765556911
Customer 38 arrived: 48.860039698791745
Customer 39 arrived: 49.36715413961523
Customer 37 exited service: 49.58351890453754
Customer 38 entered service: 49.58351890453754
Customer 40 arrived: 50.35826885149527
Customer 36 exited service: 52.403918124049895
Customer 39 entered service: 52.403918124049895
Customer 41 arrived: 52.96197822449703
Customer 38 exited service: 52.990352268380526
Customer 40 entered service: 52.990352268380526
```

```
Customer 40 exited service: 53.82165508751175
Customer 41 entered service: 53.82165508751175
Customer 41 exited service: 54.33122247322352
Customer 42 arrived: 54.56609776792658
Customer 42 entered service: 54.56609776792658
Customer 43 arrived: 54.970819902879676
Customer 42 exited service: 55.18640899636045
Customer 43 entered service: 55.18640899636045
Customer 44 arrived: 56.365628688018994
Customer 39 exited service: 56.50247150760596
Customer 44 entered service: 56.50247150760596
Customer 44 exited service: 57.208195848015876
Customer 45 arrived: 58.043182063739444
Customer 45 entered service: 58.043182063739444
Customer 45 exited service: 58.31192803415694
Customer 43 exited service: 59.72870455150809
Customer 46 arrived: 60.053590672544495
Customer 46 entered service: 60.053590672544495
Customer 47 arrived: 62.014091189412945
Customer 47 entered service: 62.014091189412945
Customer 48 arrived: 62.89373798824384
Customer 49 arrived: 64.48911558625039
Customer 47 exited service: 65.09550251187065
Customer 48 entered service: 65.09550251187065
Customer 48 exited service: 65.94336896105845
Customer 49 entered service: 65.94336896105845
Customer 49 exited service: 66.00215658802487
Customer 46 exited service: 66.25365488979618
Customer 50 arrived: 71.6394745738439
```

```
Customer 50 entered service: 71.6394745738439
Customer 51 arrived: 73.48105063571381
Customer 51 entered service: 73.48105063571381
Customer 52 arrived: 75.13416295030062
Customer 53 arrived: 75.43915175120722
Customer 50 exited service: 76.54258122608648
Customer 52 entered service: 76.54258122608648
Customer 51 exited service: 76.63524970489402
Customer 53 entered service: 76.63524970489402
Customer 54 arrived: 76.8456475311067
Customer 52 exited service: 78.69225749181444
Customer 54 entered service: 78.69225749181444
Customer 53 exited service: 80.29396982461465
Customer 54 exited service: 81.35590765303633
Customer 55 arrived: 84.43925400786016
Customer 55 entered service: 84.43925400786016
Customer 56 arrived: 84.54046538023812
Customer 56 entered service: 84.54046538023812
Customer 57 arrived: 85.43671085657867
Customer 56 exited service: 85.43731554023604
Customer 57 entered service: 85.43731554023604
Customer 57 exited service: 86.85950975658587
Customer 58 arrived: 88.75431647681815
Customer 58 entered service: 88.75431647681815
Customer 59 arrived: 89.13345635257599
Customer 60 arrived: 89.34529285913138
Customer 55 exited service: 90.1539674354481
Customer 59 entered service: 90.1539674354481
Customer 59 exited service: 90.27287716154173
```

```
Customer 60 entered service: 90.27287716154173
Customer 61 arrived: 92.35980967835778
Customer 62 arrived: 92.62882153096564
Customer 63 arrived: 93.42754967506838
Customer 64 arrived: 93.90866461206957
Customer 60 exited service: 93.91094142440427
Customer 61 entered service: 93.91094142440427
Customer 65 arrived: 94.23276195823995
Customer 61 exited service: 94.67748717670953
Customer 62 entered service: 94.67748717670953
Customer 58 exited service: 95.2130778338892
Customer 63 entered service: 95.2130778338892
Customer 66 arrived: 96.16702984650058
Customer 63 exited service: 96.2291084752117
Customer 64 entered service: 96.2291084752117
Customer 67 arrived: 96.5560961327119
Customer 68 arrived: 96.8923520036099
Customer 64 exited service: 97.9398196600904
Customer 65 entered service: 97.9398196600904
Customer 69 arrived: 98.08876651654529
Customer 70 arrived: 98.13494747971596
Customer 71 arrived: 99.14853418525063
Customer 65 exited service: 100.39056577391867
Customer 66 entered service: 100.39056577391867
Customer 72 arrived: 101.42016715937558
Customer 66 exited service: 101.44400165655772
Customer 67 entered service: 101.44400165655772
Customer 73 arrived: 102.27934723024107
Customer 67 exited service: 102.46094411928101
```

```
Customer 68 entered service: 102.46094411928101
Customer 74 arrived: 102.47635628895546
Customer 62 exited service: 102.48625981633232
Customer 69 entered service: 102.48625981633232
Customer 75 arrived: 102.83498163616316
Customer 68 exited service: 102.84173630012062
Customer 70 entered service: 102.84173630012062
Customer 76 arrived: 103.24590422882913
Customer 69 exited service: 103.46264559955941
Customer 71 entered service: 103.46264559955941
Customer 70 exited service: 104.00084411447494
Customer 72 entered service: 104.00084411447494
Customer 72 exited service: 104.28117492719811
Customer 73 entered service: 104.28117492719811
Customer 71 exited service: 104.88532041789259
Customer 74 entered service: 104.88532041789259
Customer 73 exited service: 105.15894497050043
Customer 75 entered service: 105.15894497050043
Customer 75 exited service: 105.371729061353
Customer 76 entered service: 105.371729061353
Customer 76 exited service: 105.66368697371337
Customer 77 arrived: 105.70610449261844
Customer 77 entered service: 105.70610449261844
Customer 78 arrived: 105.71982272656982
Customer 79 arrived: 105.72112288977493
Customer 77 exited service: 106.26948585938383
Customer 78 entered service: 106.26948585938383
Customer 74 exited service: 106.54679601734091
Customer 79 entered service: 106.54679601734091
```

```
Customer 80 arrived: 106.90267896774559
Customer 81 arrived: 107.78743333649001
Customer 82 arrived: 107.84262515816941
Customer 79 exited service: 107.8900250323206
Customer 80 entered service: 107.8900250323206
Customer 80 exited service: 108.30995212742235
Customer 81 entered service: 108.30995212742235
Customer 83 arrived: 109.05298426191241
Customer 81 exited service: 109.42097697182021
Customer 82 entered service: 109.42097697182021
Customer 82 exited service: 109.46539565493195
Customer 83 entered service: 109.46539565493195
Customer 84 arrived: 109.60471324342876
Customer 85 arrived: 109.72300600276935
Customer 78 exited service: 109.75589997073865
Customer 84 entered service: 109.75589997073865
Customer 86 arrived: 112.25383464141761
Customer 83 exited service: 112.86263678628694
Customer 85 entered service: 112.86263678628694
Customer 84 exited service: 113.23355969959998
Customer 86 entered service: 113.23355969959998
Customer 86 exited service: 114.04793464615837
Customer 87 arrived: 114.22038694416149
Customer 87 entered service: 114.22038694416149
Customer 88 arrived: 114.24421830744156
Customer 89 arrived: 114.91417302193354
Customer 87 exited service: 115.50748997801637
Customer 88 entered service: 115.50748997801637
Customer 88 exited service: 115.52834517630845
```

```
Customer 89 entered service: 115.52834517630845
Customer 90 arrived: 115.60744575510695
Customer 85 exited service: 115.67714942777363
Customer 90 entered service: 115.67714942777363
Customer 91 arrived: 116.08562550397437
Customer 92 arrived: 116.32372865155169
Customer 89 exited service: 117.2150449878906
Customer 91 entered service: 117.2150449878906
Customer 93 arrived: 118.58928482329404
Customer 91 exited service: 119.00016251605943
Customer 92 entered service: 119.00016251605943
Customer 92 exited service: 119.16749987653277
Customer 93 entered service: 119.16749987653277
Customer 93 exited service: 119.88100153740699
Customer 94 arrived: 120.2982317563385
Customer 94 entered service: 120.2982317563385
Customer 95 arrived: 120.85732149555938
Customer 94 exited service: 120.95959252048127
Customer 95 entered service: 120.95959252048127
Customer 95 exited service: 121.1510969105177
Customer 96 arrived: 122.80449723968039
Customer 96 entered service: 122.80449723968039
Customer 96 exited service: 123.41502232443476
Customer 97 arrived: 123.70432765275488
Customer 97 entered service: 123.70432765275488
Customer 97 exited service: 125.65171039157883
Customer 98 arrived: 126.62995559203662
Customer 98 entered service: 126.62995559203662
Customer 98 exited service: 126.96625414992693
```

```
Customer 99 arrived: 127.55807931536204
Customer 99 entered service: 127.55807931536204
Customer 100 arrived: 128.29984972160915
Customer 90 exited service: 128.3246246899848
Customer 100 entered service: 128.3246246899848
Customer 100 exited service: 129.00038699065178
Customer 99 exited service: 131.05926704873303
```

5.4 Параметрическое исследование: сканирование λ

```
lambdas = [0.5, 0.9, 1.2, 1.5]
results = []

for lam_val in lambdas
    rng_local = StableRNG(42)
    arrival_dist_local = Exponential(1 / lam_val)
    service_dist_local = Exponential(1 / mu)
    sim = Simulation()
    server = Resource(sim, num_servers)
    arrival_time = 0.0
    service_times = Float64[]
    wait_times = Float64[]
    for i in 1:num_customers
      arrival_time += rand(rng_local, arrival_dist_local)
       service_time = rand(rng_local, service_dist_local)
        push!(service_times, service_time)
     push!(wait_times, max(0.0, service_time - (arrival_time / num_cu
```

```
    end
    rho = lam_val / (num_servers * mu)
    push!(results, (
        lambda = lam_val,
        rho = rho,
        mean_wait = mean(wait_times),
        mean_service = mean(service_times)
    ))
    println("□ = $lam_val, □ = $(round(rho, digits=3))")
end

df = DataFrame(results)
CSV.write(datadir(script_name, "mmc_scan.csv"), df)
```

```
□ = 0.5, □ = 0.5
□ = 0.9, □ = 0.9
□ = 1.2, □ = 1.2
□ = 1.5, □ = 1.5

"/home/dabakharev/work/study/2026-1/2026-1==study-simulation-modeli
```

5.5 График

```
p = plot(df.lambda, [df.mean_wait df.mean_service],
    label=["□□□□□□□ □□□□□□□□" "□□□□□□□ □□□□□□□□□□□□"],
    marker=:circle,
    xlabel="□ (□□□□□□□□□□□□ □□□□□□□□)",
    ylabel="□□□□□",
```

```
    title="M/M/$(num_servers): □□□□□□□□□□□□ □□ □")
savefig(plotsdir(script_name, "mmc_scan.png"))

println("\n□□□□□□□□□□□□□□ □ □□□□□□□□□□.")
println(df)
```

```
□□□□□□□□□□□□□□ □ □□□□□□□□□□.
4×4 DataFrame
 Row □ lambda   rho       mean_wait  mean_service
     □ Float64  Float64   Float64    Float64
─────┼──────────────────────────────────────────────
   1 □      0.5       0.5    1.33167        2.10658
   2 □      0.9       0.9    1.61528        2.10658
   3 □      1.2       1.2    1.72134        2.10658
   4 □      1.5       1.5    1.79081        2.10658
```

6 Модель Росса

Система с N работающими машинами, S запасными и одним ремонтником. Система падает, когда резерв исчерпан и очередная машина ломается.

```
import Pkg
Pkg.activate("/home/dabakharev/work/study/2026-1/2026-1==study-simu
import Pkg
Pkg.activate("/home/dabakharev/work/study/2026-1/2026-1==study-simu
using DrWatson

using ResumableFunctions
using ConcurrentSim
using Distributions
using Random
using StableRNGs
using DataFrames, CSV, Plots, Statistics

ENV["GKSwstype"] = "100"

script_name = "ross_model"
mkpath(plotsdir(script_name))
mkpath(datadir(script_name))
```

```
  Activating project at `~/work/study/2026-1/2026-1==study-simulation
```

```
  Activating project at `~/work/study/2026-1/2026-1==study-simulation
```

"/home/dabakharev/work/study/2026-1/2026-1==study-simulation-modeli

6.1 Параметры (уменьшены для скорости)

```
const RUNS   = 2          # [illegible] [illegible] [illegible] [illegible] — [illegible]
const N      = 10          # 5 [illegible] [illegible] ([illegible] [illegible])
const SEED   = 150
const LAMBDA = 100.0
const MU     = 1.0
```

```
1.0
```

6.2 Машина

```
@resumable function machine(env::Environment, repair_facility::Resou
                            spares::Store{Process}, rng)
    while true
        try
            @yield timeout(env, Inf)
        catch
        end
      @yield timeout(env, rand(rng, Exponential(LAMBDA)))

        get_spare = take!(spares)
        @yield get_spare | timeout(env)
```

```
        if state(get_spare) != ConcurrentSim.idle
            @yield interrupt(value(get_spare))
        else
            throw(StopSimulation("No more spares!"))
        end

        @yield request(repair_facility)
        @yield timeout(env, rand(rng, Exponential(MU)))
        @yield unlock(repair_facility)
        @yield put!(spares, active_process(env))
    end
end
```

```
machine (generic function with 1 method)
```

6.3 Запуск: N основных + S запасных

```
@resumable function start_sim(env::Environment, repair_facility::Res
                              spares::Store{Process}, N, S, rng)
    for i in 1:N
     proc = @process machine(env, repair_facility, spares, rng)
        @yield interrupt(proc)
    end
    for i in 1:S
     proc = @process machine(env, repair_facility, spares, rng)
        @yield put!(spares, proc)
    end
end
```

```
start_sim (generic function with 1 method)
```

6.4 Один прогон

```
function sim_repair(; N=N, S=3, seed=SEED)
    rng = StableRNG(seed)
    sim = Simulation()
    repair_facility = Resource(sim)
    spares = Store{Process}(sim)

   @process start_sim(sim, repair_facility, spares, N, S, rng)

    try
        run(sim)
    catch e
        if !isa(e, StopSimulation)
            rethrow(e)
        end
    end
    return now(sim)
end
```

```
sim_repair (generic function with 1 method)
```

6.5 Базовый прогон

```
println("=== [illegible] [illegible] ===")
println("N = $N, S = 3, LAMBDA = $LAMBDA, MU = $MU")

results = Float64[]
for i in 1:RUNS
    push!(results, sim_repair(S=3, seed=SEED + i))
end
println("Crash time (S=3): ", round(mean(results), digits=2))
```

```
=== [illegible] [illegible] ===
N = 10, S = 3, LAMBDA = 100.0, MU = 1.0
Crash time (S=3): 12802.25
```

6.6 Сканирование по S

```
println("\n=== [illegible] S ===")
S_range = [1, 2, 3, 5]
scan_results = []
for s_val in S_range
    rs = Float64[]
    for i in 1:RUNS
        push!(rs, sim_repair(S=s_val, seed=SEED + i))
    end
    avg_s = mean(rs)
    push!(scan_results, (S=s_val, crash_time=avg_s))
  println("S = $s_val, crash time = $(round(avg_s, digits=2))")
end
```

```
df = DataFrame(scan_results)
CSV.write(datadir(script_name, "ross_scan.csv"), df)
```

```
=== □□□□□□□□□□□□ S ===
S = 1, crash time = 20.77
S = 2, crash time = 1096.07
S = 3, crash time = 12802.25
S = 5, crash time = 840855.82

"/home/dabakharev/work/study/2026-1/2026-1==study-simulation-modeli
```

6.7 График

```
p = plot(df.S, df.crash_time,
    marker=:circle,
    xlabel="□□□□□ □□□□□□□□ □□□□□ S",
    ylabel="□□□□□□□ □□□□□ □□ □□□□□□",
    title="□□□□□□ □□□□□: □□□□□□□□□□ □□□□□□□ □□ □□□□□□ □□ S")
savefig(plotsdir(script_name, "ross_scan.png"))

println("\n□□□□□□! □□□□□□□□□□: ", datadir(script_name, "ross_scan.csv"))
```

```
□□□□□□! □□□□□□□□□□: /home/dabakharev/work/study/2026-1/2026-1==study-si
```

7 Заключение

В ходе лабораторной работы №7:

- Освоен пакет **`ConcurrentSim.jl`** для дискретно-событийного моделирования.
- Реализована модель **M/M/c** (многоканальная СМО).
- Реализована модель **Росса** (система с резервом и ремонтом).
- Проведены параметрические исследования:
  - для M/M/c — сканирование λ;
  - для модели Росса — сканирование S.
- Показано, что **резервирование** резко увеличивает время до отказа системы.
- Сгенерированы `.jl`, `.qmd`, `.ipynb` и собран отчёт.

Список литературы